UNIVERSITY OF UYO
DEPARTMENT OF MATHEMATICS
FIRST SEMESTER EXAMINATION 2025/2026 SESSION
MTH 412: PARTIAL DIFFERENTIAL EQUATIONS TIME: 2 HOURS

INSTRUCTIONS: ATTEMPT QUESTION 1 AND ANY OTHER 3 QUESTIONS ONLY

In the equations, take $p = \frac{\partial z}{\partial x}$ and $q = \frac{\partial z}{\partial y}$

1.

a) Integrate $xy^2dx + x^2ydy + 2zdz = 0$

b) Solve the differential equation $xdx - ydy = zx^2dz$

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c) $yzdx + 2xzdy - 3xydz = 0$

2. Find the characteristics and the corresponding transport equations of the system

$$xyu_x - v_y = 0$$
$$xu_y - yv_x = 0$$

3. Show that the equations $xp = yq, z(xp + yq) = 2xy$ are compatible and solve them.

4. Find the integral curves of the following systems:

a) $\frac{dx}{1} = \frac{dy}{3} = \frac{dz}{5z+\tan(y-3x)}$

b) $\frac{dx}{yz} = \frac{dy}{xz} = \frac{dz}{xy}$

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c) $\frac{dx}{x(y-z)} = \frac{dy}{y(z-x)} = \frac{dz}{z(x-y)}$

5. Find the general integral of the following linear partial differential equations:

a) $xp + yq = z$

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b) $z(xp - yq) = y^2 - x^2$

c) $yzp + zxq = xy$

6. Use Charpit's method to obtain a complete integral for the equation

$$p^2q^2 + x^2y^2 = x^2q^2(x^2 + y^2)$$